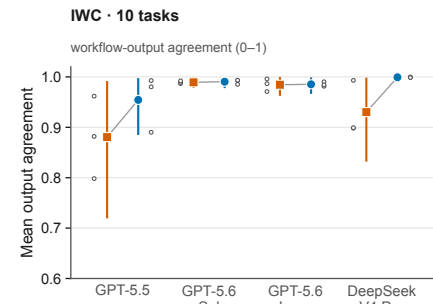
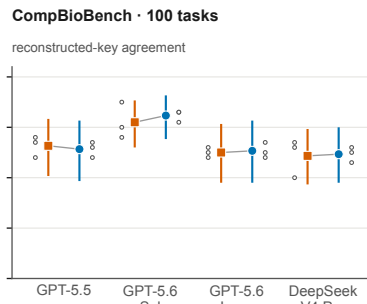
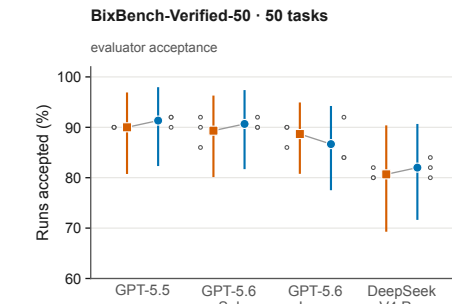
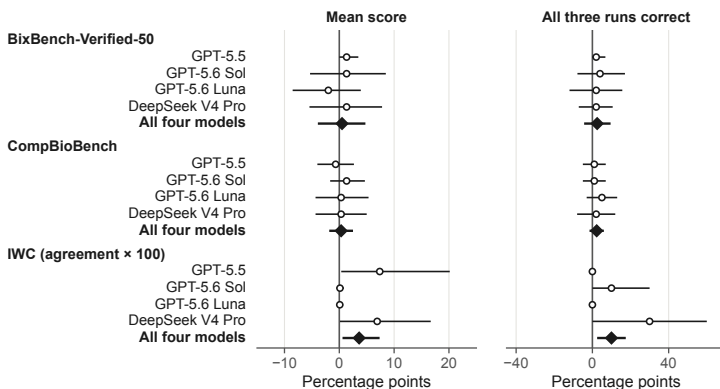




b Paired differences: Galaxy minus custom code

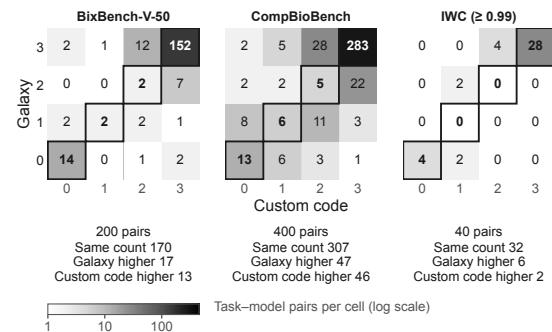
← custom code higher · Galaxy higher → (95% intervals)

Without the 94 runs that reached benchmark answers: BixBench-Verified-50 +0.6, CompBioBench +0.2 points



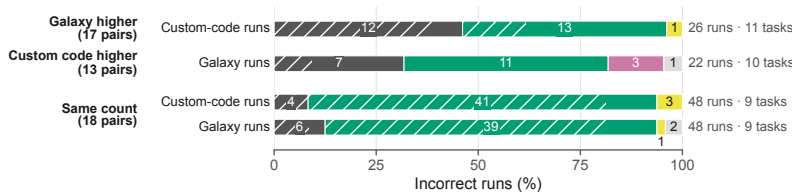
c Correct runs of three, same task and model

Outlined diagonal, same count; above it, Galaxy had more correct runs; below it, custom code had more



d Why the conditions disagree (BixBench-Verified-50)

Incorrect runs by primary cause (AI-assisted audit), for task-model pairs whose conditions differ or agree
An independent second rater agreed on the cause for 35 of 45 sampled runs ($\kappa = 0.56$)



Traced case: bix-45-q1 (relative composition variability)

	Custom code	Galaxy
GPT-5.5	○ ○ ○	○ ○ ○
GPT-5.6 Sol	● ● ●	○ ○ ○
GPT-5.6 Luna	● ○ ●	○ ○ ○
DeepSeek V4 Pro	● ● ●	○ ○ ○

8 of 12 correct

0 of 12 correct

Successful custom-code runs matched the PhyKIT version to the date of the input archive. The Galaxy PhyKIT wrapper, like current PhyKIT, returned a different value, and the wrapper version was not shown to the agent; the reference encodes the older value.

● correct ○ incorrect (one run each)

No answer validation
 Benchmark specification or scoring
 Lacking biological knowledge
 Not able to use Galaxy
 No answer submitted
 Validation and specification both implicated